

WASTE SEGREGATION USING DEEP LEARNING

Dibya Ranjan Dash (Roll Number: 23035010567)

Term Project Report
Trimester 8

B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
dibya.dash@op.iitg.ac.in

ABSTRACT

This project presents an AI-based garbage classification system designed to automatically categorize waste into different classes such as cardboard, glass, metal, organic, paper, plastic, and trash. The problem of improper waste segregation is significant as it negatively impacts environmental sustainability and reduces the efficiency of recycling processes.

To address this challenge, a deep learning-based approach was adopted using a Convolutional Neural Network (CNN) built on the MobileNetV2 architecture. The model was trained on an image dataset with augmentation techniques such as rotation, zooming, and flipping to enhance generalization. Additionally, a Flask-based API was developed to serve predictions, and a Streamlit dashboard was implemented to provide an interactive interface for users.

Experimental results indicate that the model can accurately classify waste images with high confidence for most categories. The system demonstrates the practical feasibility of using AI for automated waste segregation and highlights its potential in supporting smart waste management and sustainable urban development.

Index Terms — Deep Learning, Image Classification, MobileNetV2, Waste Segregation, Computer Vision

1. INTRODUCTION

Waste management is a growing global challenge driven by increasing urbanization, population growth, and consumer activity. Improper disposal and inefficient segregation of waste contribute significantly to environmental pollution, loss of recyclable resources, and increased pressure on landfills. Traditional manual sorting methods are labor-intensive, time-consuming, and prone to human error, making them inefficient for large-scale waste management systems. These challenges highlight the urgent need for automated and intelligent solutions.

Artificial Intelligence (AI), particularly Computer Vision, offers a powerful approach to automate waste classification. By analyzing visual features such as texture, shape, and color patterns, machine learning models can distinguish between different types of materials, including plastic, glass, paper, and organic waste. This enables faster and more accurate waste segregation, which is essential for improving recycling efficiency and supporting sustainable environmental practices.

In this project, an AI-based garbage classification system is developed using a pre-trained MobileNetV2 model, which leverages transfer learning to achieve efficient and accurate classification. The

system is designed as an end-to-end solution comprising multiple components. It includes:

- A robust training pipeline for model development using augmented image datasets.
- A Flask-based API for handling real-time prediction requests.
- A Streamlit dashboard that provides an interactive and user-friendly interface for visualization and analysis.

Furthermore, the system incorporates features such as confidence-based predictions, recycling advice recommendations, and real-time visualization of classification results. These additions enhance usability and make the system suitable for practical deployment in smart waste management scenarios. Overall, this project demonstrates how AI-driven solutions can play a crucial role in addressing real-world environmental challenges and advancing sustainable development.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

The goal of this project is to develop an automated system capable of classifying garbage images into predefined categories for effective waste segregation. The proposed system takes an image of waste as input and performs classification into categories such as cardboard, glass, metal, organic, paper, plastic, and trash. In addition to classification, the system provides a confidence score indicating prediction reliability and suggests appropriate recycling advice for each category.

The problem operates under the constraint that input images may vary significantly in quality, lighting conditions, background complexity, and object orientation. Additionally, certain waste categories, such as paper and cardboard, exhibit high visual similarity, increasing the difficulty of accurate classification. The system assumes that input data is sourced from real-world environments, where such variations are inevitable, requiring the model to generalize effectively across diverse conditions.

This problem is highly relevant in real-world scenarios, as improper waste segregation leads to reduced recycling efficiency and increased environmental pollution. An automated classification system can support sustainable waste management by enabling faster, more accurate sorting of materials, thereby contributing to improved recycling processes and the development of smart city solutions.

2.2. Objectives

The main objectives of this project are as follows:

1. To study existing deep learning approaches for image classification tasks.
2. To design and implement a waste classification system using the MobileNetV2 model.
3. To preprocess and augment image datasets to improve model generalization and accuracy.
4. To develop a RESTful API using Flask to handle prediction requests.
5. To build an interactive user interface using Streamlit for real-time classification.
6. To evaluate the model performance using appropriate metrics such as accuracy and prediction confidence.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

The overall system follows a structured pipeline consisting of the following stages:

1. Data Loading & Augmentation

Images are loaded from a directory-based dataset where each folder represents a specific waste class. To improve the robustness of the model, data augmentation techniques such as rotation, zooming, and horizontal flipping are applied. The dataset is then split into training and validation sets in an 80:20 ratio.

2. Model Training

The model is built using MobileNetV2 with transfer learning. The pre-trained layers are frozen to retain their learned features, and a fully connected classification layer is added to adapt the model to the waste classification task. This approach reduces training time while maintaining performance.

3. Model Saving

After training, the model is saved as `waste_model.h5`. The mapping between class labels and indices is stored in `class_indices.json` to ensure consistency during prediction.

4. Prediction System

A Flask-based API is developed to handle user input. The API accepts image requests, processes them, and returns the predicted class along with the confidence score.

5. User Interface

A Streamlit dashboard is implemented to provide an interactive interface. Users can upload images, view predictions, and analyze results through charts and visual elements.

3.2. Technical Details

1. Data Preprocessing

All images are resized to 224×224 pixels to match the input requirements of the model. Pixel values are normalized to the range $[0,1]$ to ensure stable training. Augmentation techniques such as horizontal flipping, rotation (up to 25°), and zooming (up to 20%) are applied to improve model generalization.

2. Model Architecture

The system uses MobileNetV2, a pre-trained convolutional neural network. The architecture is extended by adding a Global Average Pooling layer followed by a Dense layer with Softmax activation to perform multi-class classification.

3. Training Configuration

The model is trained using the Adam optimizer and categorical crossentropy as the loss function. Training is performed for 15 epochs with a batch size of 32, balancing efficiency and performance.

4. Prediction Logic

Input images are processed using OpenCV, resized, and normalized before being passed to the trained model. The model outputs a probability distribution over all classes, and the class with the highest probability is selected as the final prediction.

5. Additional Features

The system includes a recycling advice module that provides category-specific disposal suggestions. It also supports real-time statistics tracking and offers interactive data visualization through the Streamlit dashboard.

4. EXPERIMENTS AND RESULTS

This section presents the experimental setup and obtained results.

4.1. Experimental Setup

The experimental setup for the proposed AI-based garbage classification system was designed to evaluate the performance of the deep learning model under realistic conditions. The dataset used in this project consists of a garbage classification image dataset containing labeled images of waste materials. The dataset includes seven distinct categories, namely cardboard, glass, metal, organic, paper, plastic, and trash. These categories represent commonly encountered types of waste and are essential for effective segregation.

The system was implemented using Python as the primary programming language, along with TensorFlow as the deep learning framework for model development and training. For deployment and interaction, Flask was used to build a REST API for handling prediction requests, while Streamlit was utilized to develop an interactive web-based dashboard for visualization and user interaction.

The performance of the model was evaluated using standard metrics such as classification accuracy and prediction confidence. Accuracy measures the overall correctness of the model's predictions, while prediction confidence indicates the probability associated with each predicted class, providing insight into the reliability of the model's outputs.

4.2. Results

The trained model successfully classifies garbage images into seven predefined categories. It achieves high confidence scores for visually distinct materials such as glass, plastic, organic, and cardboard. However, moderate misclassification is observed between visually similar categories, particularly paper and cardboard.

The model shows consistent performance on clear images with distinct features, indicating effective feature extraction through transfer learning. Prediction confidence values further help in understanding the reliability of outputs, especially in borderline cases. Overall, the system demonstrates promising results and has potential for real-world deployment in automated waste management systems.

Sample Outputs

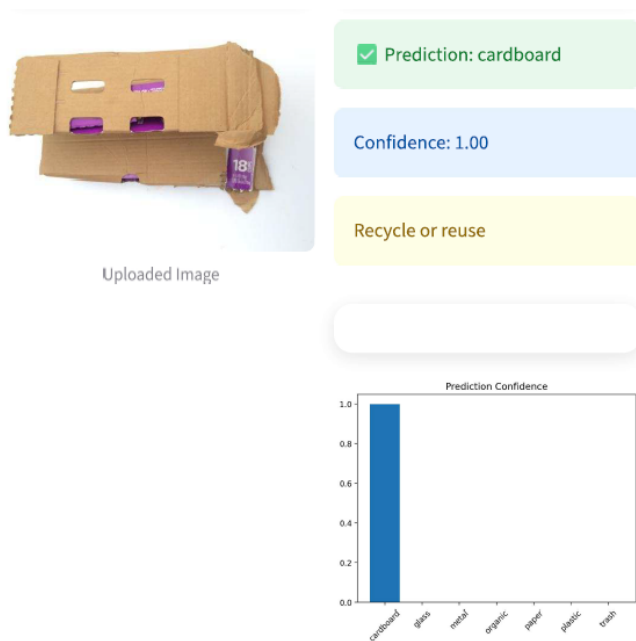


Fig. 1. System correctly classifies a cardboard object with high confidence

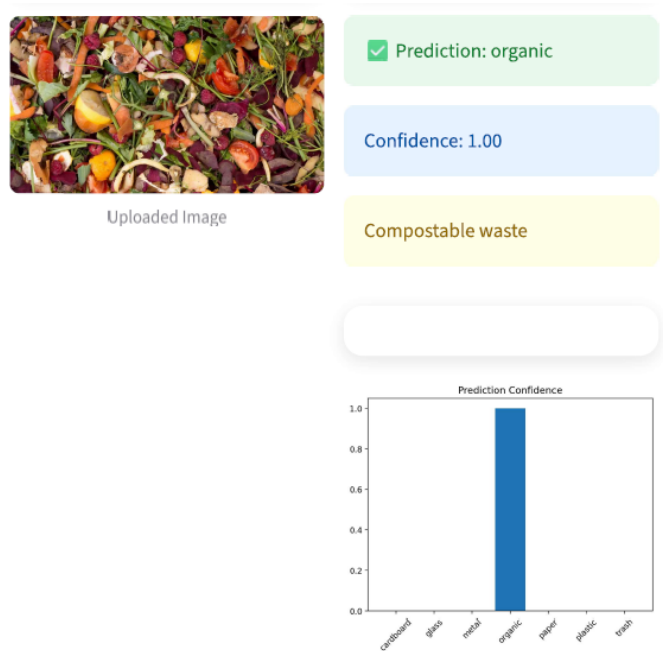


Fig. 3. Model identifies organic waste and recommends composting

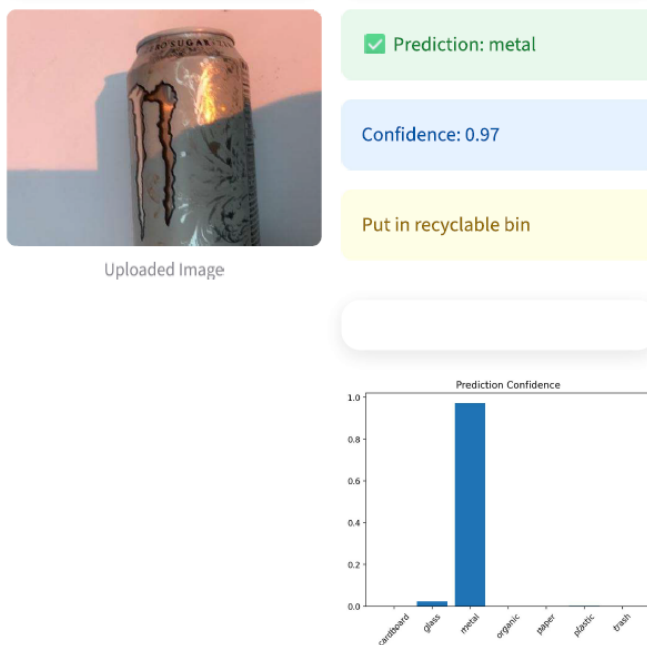


Fig. 2. System detects a metal object and suggests recycling

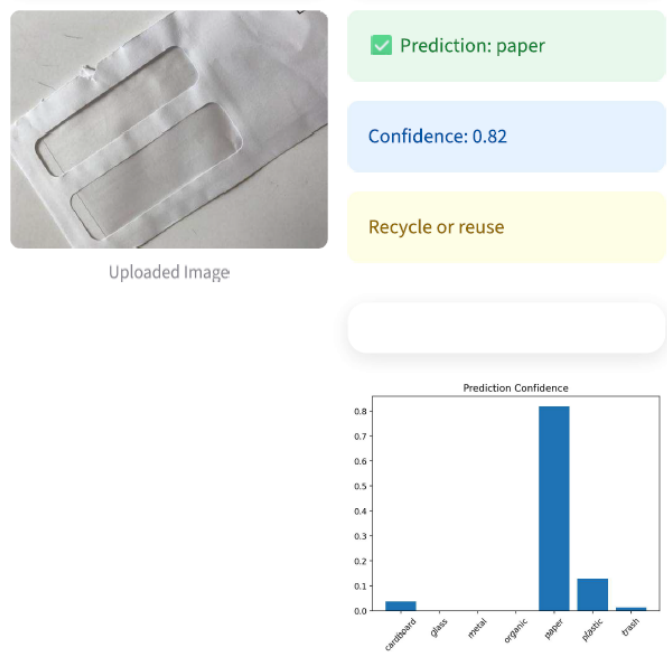


Fig. 4. System correctly classifies a paper object with moderate confidence

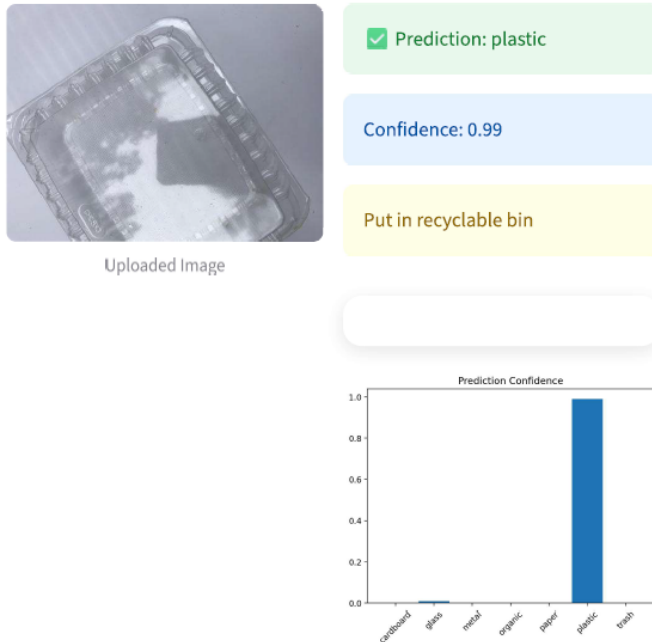


Fig. 5. Model identifies a plastic object with high confidence and recommends recycling

5. DISCUSSION

The proposed system demonstrates strong performance when processing structured and high-quality images. The use of transfer learning with MobileNetV2 significantly reduces training time while maintaining reasonable classification accuracy, making it a suitable choice for resource-efficient deployment.

Observations

- Data augmentation techniques such as rotation, zooming, and flipping improve the model's ability to generalize across varying image conditions.
- The model performs better when trained on a balanced dataset with sufficient representation of all classes.
- Background noise and clutter in images can slightly affect prediction accuracy.

Challenges

- Misclassification occurs between visually similar categories, such as paper and cardboard.
- Limited dataset diversity restricts the model's ability to generalize to unseen real-world scenarios.
- Model performance is sensitive to image clarity, lighting, and input quality.

Trade-offs

- Faster training time is achieved at the cost of slightly reduced accuracy compared to deeper and more complex architectures.
- The use of a lightweight model like MobileNetV2 ensures efficiency, but may limit performance compared to heavier models such as ResNet or EfficientNet.

Additionally, the system demonstrates good scalability and can be extended to include more waste categories with minimal architectural changes. The integration of a user-friendly Streamlit dashboard

enhances accessibility, allowing even non-technical users to interact with the model effectively. The real-time prediction capability further highlights its practical usability in smart waste management systems. With improved dataset diversity, better data quality, and fine-tuning of model parameters, the overall system performance can be significantly enhanced for real-world deployment.

6. CONCLUSION AND FUTURE WORK

This project successfully demonstrates an AI-powered garbage classification system using deep learning techniques. The integration of a trained model with a Flask-based API and a Streamlit dashboard makes the solution both practical and interactive. The system highlights the potential of artificial intelligence in automating waste segregation, thereby contributing to efficient recycling and sustainable environmental practices.

Key Contributions

- Development of an automated waste classification system using deep learning.
- Design of an interactive dashboard for enhanced user engagement and visualization.
- Implementation of real-time prediction with confidence scores and recycling advice.

Future Work

- Improve classification accuracy by using advanced architectures such as EfficientNet or ResNet.
- Expand the dataset size and diversity to enhance generalization capability.
- Deploy the system as a mobile or web application for wider accessibility.
- Integrate IoT-enabled smart bins for real-time automated waste segregation in practical scenarios.

7. ARTIFACTS AND DEMONSTRATIONS

This section provides external resources that support the implementation and experimental results presented in this report. All artifacts are designed to ensure reproducibility and transparency of the project work.

7.1. Source Code Repository

The complete source code includes Python implementations for model training (TensorFlow), a Flask API for real-time predictions, and a Streamlit dashboard for visualization.

Link: <https://github.com/dibyardash/Waste-Segregation-git>

7.2. Video Demonstration

A short video demonstration showcasing the project structure, working pipeline, and key outputs such as image classification, prediction confidence, and dashboard interaction is provided.

Link: <https://youtu.be/Sij1Q2UYKzg>

8. REFERENCES

1. TensorFlow Documentation, Available at: <https://www.tensorflow.org/>
2. Python Software Foundation, Python Language Reference, Version 3.12, 2026. Available at: <https://www.python.org/>
3. OpenCV Documentation, Available at: <https://docs.opencv.org/>
4. Streamlit Documentation, Available at: <https://docs.streamlit.io/>
5. Flask Documentation, Available at: <https://flask.palletsprojects.com/>
6. Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen, "Inverted Residuals and Linear Bottlenecks: Mobile Networks for Classification, Detection and Segmentation," Google Research, 2018.